ASIM ALTAF

Telecom Systems Engineer — SS7/MAP Signalling · HLR/HSS · eSIM (SM-DP+) · 5G Core

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PROFILE

I build the signalling and subscriber-data systems mobile networks run on. At Safarifone, a mobile operator, I work on the MAP/HLR layer, a full-lifecycle eSIM provisioning platform running against a live SM-DP+, and a 3GPP-compliant UDM/UDR — the 5G successor to the HSS — in production at multi-million subscriber scale. Before that, nearly two years building VoIP/SIP infrastructure in Go that held up at 10,000+ calls per second. Go and C are my daily languages.

EXPERIENCE

Backend / Systems Engineer · **Safarifone (mobile operator)** — Nov 2025 – Present · Islamabad

Work across four production teams: SS7 signalling, subscriber data (UDM/UDR), eSIM provisioning, and the subscriber platform.

- **SS7 & signalling:** built the HLR MAP adapter (SendRoutingInfo, UpdateLocation, InsertSubscriberData) and the TCAP transaction/session manager for USSD over SIGTRAN, in Go. Wrote the SS7-to-JSON translation layer that application teams consume, plus a custom 12-byte big-endian binary framing protocol with zero-allocation parsing.
- **Subscriber data (HSS lineage):** implemented UDM/UDR to 3GPP TS 29.503 / 29.505 / 23.502 in C with libuv — SUPI/SUCI handling and authentication vector generation. The same job an HSS does, one generation newer. Replaced a thread-heavy design with an async event loop; p95/p99 latency dropped measurably. Valgrind profiling across a 50K+ line C codebase.
- **eSIM:** architected our ES2+ provisioning service in Go against a live SM-DP+ (GSMA SGP.02): DownloadOrder through GetProfileStatus, mTLS client auth, callback handling, the full profile state machine, and HLR lookups for IMSI/ICCID/SQN resolution.
- Day to day: Go, C/C++, gRPC/Protobuf, Kubernetes, Docker, Temporal.io, Oracle, PostgreSQL, Redis.

Golang Developer (part-time remote contract) · **ContactVA — US enterprise clients** — Sep 2024 – Jul 2026

- Built VoIP/SIP infrastructure in Go: a UDP signalling server that held up at 10,000+ calls per second, with SIP signalling, RTP relay and OpenSIPS call routing.
- Cut API latency from ~1,000 ms to ~300 ms by reworking OpenSIPS flows and the PostgreSQL/Redis layer.
- *Ran alongside my full-time role from late 2025, with the client's agreement, until the contract wrapped in July 2026.*

Backend Developer · **ZIXEL** — Sep 2023 – Jun 2024 · Islamabad

- First role out of university. Built and maintained Go microservices across five products, added caching and autoscaling that cut response times roughly 20%, and mentored two junior backend developers.

TECHNICAL SCOPE

- **Signalling & core:** SS7 (MAP, TCAP, USSD, SIGTRAN), HLR/HSS integration, 5G SBA (UDM/UDR, TS 29.503/29.505/23.502), SIP/RTP
- **eSIM / RSP:** GSMA SGP.02, ES2+, SM-DP+ integration, profile lifecycle, mTLS; familiar with SGP.22 consumer flows
- **Languages & infra:** Go, C, C++ (libuv), gRPC, Protobuf, Kubernetes, Docker, Oracle, PostgreSQL, Redis, Temporal.io

EDUCATION

BS Computer Science — National University of Computer & Emerging Sciences (FAST-NUCES) — 2019 – 2023